

1.11

$$\sigma(x) = \frac{1}{1+e^x}$$

$$x_{t+1} = f(x_t, \omega, \theta) = \omega \cdot \sigma(x_t) + \theta$$

fixed pts. $x_{t+1}^* = x_t^* = x^*$

$$\frac{\omega}{1+e^{-x^*}} + \theta = x^*$$

$$\omega + \theta + e^{-x^*} = x^* + x^* e^{-x^*}$$

$$\omega + \theta = x^* + e^{-x^*} (x^* - 1)$$

$$\omega + \theta = x^* + x^* e^{-x^*} - e^{-x^*}$$

Numerical.

2. Logistic Map

$$x_{n+1} = f(x_n) = r x_n (1 - x_n)$$

1) $0 \leq x_n \leq 1$ and $0 \leq r \leq 4$, we have

$$0 \leq f(x_n) \leq 1$$

end pts, $x_n = 0 \Rightarrow 0$ ok

$x_n = 1 \Rightarrow 0$ ok

$$\max_{x_n} f(x_n) \Rightarrow f'(x_n) = r - 2rx_n = 0$$

$$x_{n_{\max}} = \frac{1}{2}$$

$$f(x_{\max}) = f\left(\frac{1}{2}\right) = r \frac{1}{2} \left(1 - \frac{1}{2}\right) = r \frac{1}{4}$$

$$\Rightarrow r_{\max} = 4 \Rightarrow f(x_{\max}) = 1$$

$$f''(x_n) = -2r < 0, \text{ so it is a max.}$$

3.

show 2 cycle for $s \geq 3$

$$F^2(x^*, s) = x^*$$

$$x_n(s-1-sx_n) = 0$$

$$x_{n+1} = sx_n(1-x_n) = sx_n - sx_n^2 = x_n \Rightarrow x_n = \frac{s-1}{s}$$

$$x_{n+2} = s(sx_n(1-x_n))(1-sx_n(1-x_n))$$

$$\Rightarrow x^* = x = \frac{(s^2x - s^2x^2)}{(1-sx + sx^2)}$$

$$x = \frac{s^2x - s^3x^2 + s^3x^3 - s^2x^2 + s^3x^3 - s^3x^4}{1-sx + sx^2}$$

$$0 = x(s^2-1) - x^2(s^2+s^3) + 2x^3s^3 - s^3x^4$$

we know $x=0$, and $x = \frac{s-1}{s}$ are roots.

$$0 = (s^2-1) - x(s+1)s^2 + 2x^2s^3 - s^3x^3 \div (x - \frac{s-1}{s})$$

polynomial division

$$0 = -x^2s^3 + s^2x + s^2x - s^2 - s$$

$$0 = -x^2s^2 + s^2x + sx - s - 1$$

$$0 = x^2s^2 - x(s+1)s + (s+1)$$

$$x = \frac{(r+1) \pm \sqrt{(r+1)^2 r^2 - 4r^2(r+1)}}{2r^2}$$

$$x = \frac{r+1}{2r} \pm \frac{\sqrt{r^2(r+1)(r+1-4)}}{2r^2}$$

$$x = \frac{r+1 \pm \sqrt{(r+1)(r-3)}}{2r} \quad \text{we call } x_{\pm} = q \text{ and } p$$

for real distinct roots: $0 < (r+1)(r-3)$

$$\Rightarrow 0 < r^2 - 2r - 3 \Rightarrow r = \frac{2 \pm \sqrt{4+12}}{2} \Rightarrow \cancel{5} \wedge r = 3$$

$r \text{ defined } \geq 0$

with $r = 3$ we have just one solution
for $f^{(2)}(x) = x$, other than $x = 0 \wedge x = \frac{r-1}{r}$

so $r > 3$ for $\exists$ of 2-cycle.

stability: for stability at q and p

for two cycle $x = p = f(f(x)) = f(f(p)), f(p) = q$

chain rule

$$\frac{d}{dx} [f(f(x))]_{x=p} = f'(f(p)) \cdot f'(p) = f'(q) \cdot f'(p)$$

similar for $x=q$

so $|f'(q) \cdot f'(p)|$ gives stability of 2 cycle.

$$|f'(q) \cdot f'(p)| < 1 \text{ for stable.}$$

> 1 unstable.

$$f(x) = rx(1-x) \Rightarrow f'(x) = r - 2rx$$

$$f'(q) \cdot f'(p) = (r - 2rp)(r - 2rq)$$

$$\Rightarrow |r^2 - 2r^2(q+p) + 4r^2qp| = 1$$

$$1 = |r^2(1 - 2(q+p) + 4qp)|$$

$$q+p = \frac{r(1 + \sqrt{(5+1)(r-3)})}{2r} + \frac{r(1 - \sqrt{(5+1)(r-3)})}{2r}$$

$$= \frac{\cancel{r}(5+1)}{\cancel{r}2} = \frac{5+1}{r}$$

$$qp = \left(\frac{r(1 + \sqrt{(5+1)(r-3)})}{2r} \right) \left(\frac{r(1 - \sqrt{(5+1)(r-3)})}{2r} \right)$$

$$Q P = \frac{(s+1)^2 - (s+1)(s-3)}{4s^2} = \frac{\cancel{s^2} + 2s + 1 - \cancel{s^2} + 2s + 3}{4s^2} = \frac{4(s+1)}{4s^2}$$

$$1 = \left| s^2 \left(1 - 2 \frac{s+1}{s} + 4 \left(\frac{s+1}{s^2} \right) \right) \right|$$

$$1 = | s^2 - 2(s+1)s + 4(s+1) |$$

$$1 = | -s^2 + 2s + 4 |$$

$$0 = | -s^2 + 2s + 4 | - 1$$

$$0 = -s^2 + 2s + 4 - 1 \quad \wedge \quad 0 = s^2 - 2s - 4 - 1$$

$$\left. \begin{aligned} s &= \frac{-2}{-2} \pm \sqrt{\frac{4+12}{2}} \\ s &= 1 \pm 2, \quad \Rightarrow s=3 \end{aligned} \right\} \quad \begin{aligned} s &= \frac{2 \pm \sqrt{4+20}}{2} \\ s &= 1 \pm \frac{2\sqrt{6}}{2} \quad \Rightarrow s=1+\sqrt{6} \end{aligned}$$

$$3 < s < 1 + \sqrt{6}$$

$$\Rightarrow 1 < | -s^2 + 2s + 4 | \Rightarrow \text{stable}$$

$$s > 1 + \sqrt{6} \Rightarrow \text{unstable}$$

